BIZEV-99: Best Practice Summary

> **Series:** BIZEV — Business Events & Funnel Analysis | **Notebook:** 99 | **Created:** March 2026 | **Last Updated:** 10/14/2026

Overview

This notebook consolidates every actionable best practice from the BIZEV series (notebooks 01 through 07) into a single, definitive reference. Each practice specifies exactly what to set, with no hedging. Use this as a checklist when instrumenting business events, building conversion funnels, and creating executive reports in Dynatrace.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
Permissions	`storage:bizevents:read`, `storage:events:read`, `storage:metrics:read`, `bizevents.ingest`
Knowledge	BIZEV-01 through BIZEV-07

1. Data Model and Schema

#	Best Practice	Recommended Setting/Value	Priority	Category
1	Use `bizevents` for business transactions	`fetch bizevents` — never use `events` or `logs` for business analytics	Critical	Data Source
2	Always populate `event.type`	Set to a reverse-domain string (e.g., `com.myapp.order.completed`) on every event	Critical	Schema
3	Always populate `event.provider`	Set to the source application or system name	Critical	Schema
4	Populate `event.category` on every event	Set to a logical grouping value (e.g., `ecommerce`, `auth`, `payments`)	Recommended	Schema
5	Include a `user_id` or `session_id` field	Required for funnel analysis and distinct-user conversion rates	Critical	Schema
6	Include a numeric `amount` field on transaction events	Store as a number, never as a string — enables `sum()`, `avg()` without casting	Critical	Schema
7	Use consistent field types across all events	Always send `amount` as double, `quantity` as long — never vary types	Recommended	Schema
8	Never confuse `bizevents` with `events`	`bizevents` = business transactions; `events` = infrastructure/operational signals	Critical	Data Source

2. Ingestion and Instrumentation

#	Best Practice	Recommended Setting/Value	Priority	Category
9	Start with OneAgent auto-capture for web apps	Configure capture rules in **Settings > Collect and Capture > Business events**	Critical	Ingestion
10	Use CloudEvents format for API ingestion	`Content-Type: application/cloudevents+json` with `specversion`, `type`, `source`, `data` fields	Critical	Ingestion
11	Use batch ingestion for high-volume scenarios	`Content-Type: application/cloudevents-batch+json` — send arrays of events in a single POST	Recommended	Ingestion

#	Best Practice	Recommended Setting/Value	Priority	Category
12	Implement retry logic with exponential backoff	Retry on `429` and `5xx`; keep each request under the documented 5 MB payload limit (no request-rate figure is published)	Recommended	Ingestion
13	Use the RUM and mobile APIs for client-side capture	`dynatrace.sendBizEvent(type, fields)` (RUM JavaScript API) or the mobile agents' equivalent; backend code without a capture rule uses the ingest API — there is no OneAgent SDK method for business events	Recommended	Ingestion
14	Use OpenPipeline for span-to-bizevent mapping	A **Business event** processor in the OpenPipeline **Data extraction** stage emits a new bizevent per matching span — no second instrumentation	Recommended	Ingestion
15	Never double-instrument	If spans already carry business data, extract it with a Business event processor (Data extraction stage) — do not add a second telemetry call in code	Critical	Ingestion
16	Verify ingestion health with a 15-minute interval timeseries	Run `makeTimeseries event_count = count(), interval:15m` daily to detect ingestion gaps	Recommended	Monitoring
17	Validate required field completeness after instrumentation	Query `countIf(isNotNull(event.type)) / total` to confirm >99% field population	Critical	Monitoring

3. Event Naming and Payload Design

#	Best Practice	Recommended Setting/Value	Priority	Category
18	Use reverse-domain notation with action verb	Format: `com.<company>.<domain>.<action>` (e.g., `com.myapp.order.created`)	Critical	Naming
19	Use all-lowercase, dot-separated names	`com.myapp.payment.processed` — never `PaymentProcessed` or `PAYMENT_PROCESSED`	Critical	Naming
20	Never put version numbers in event type names	Use payload fields for versioning — `com.myapp.order.created` stays the same, add `schema_version: 2` in payload	Critical	Naming
21	Never use meaningless identifiers as event types	`event_12345` is forbidden — use descriptive domain names	Critical	Naming

#	Best Practice	Recommended Setting/Value	Priority	Category
22	Include business identifiers in payload	`order_id`, `customer_id`, `session_id`	Critical	Payload
23	Include measurable values in payload	`amount`, `quantity`, `duration_ms`	Critical	Payload
24	Include categorical dimensions in payload	`category`, `region`, `customer_tier`	Recommended	Payload
25	Never include high-cardinality strings in payload	No full URLs, stack traces, or free-text fields — parse into structured fields	Critical	Cardinality
26	Use high-cardinality fields for filtering only, not grouping	`filter order_id == "ORD-123"` is fine; `summarize by:{order_id}` on millions of values is not	Critical	Cardinality
27	Keep `summarize by:` dimensions under 1,000 distinct values	Fields like `customer_tier` (~3 values) or `product_category` (~100 values) are ideal	Recommended	Cardinality

4. Funnel Analysis

#	Best Practice	Recommended Setting/Value	Priority	Category
28	Define explicit funnel steps as distinct event types	Each step gets its own `event.type`: `product.viewed` → `cart.updated` → `checkout.started` → `payment.processed` → `order.completed`	Critical	Funnel Design
29	Use `countIf` for funnel step counts in a single query	`summarize {step1 = countIf(event.type == "..."), step2 = countIf(...)}` — never run separate queries per step	Critical	DQL Pattern
30	Use `countDistinctApprox` for distinct-user funnels	`countDistinctApprox(if(event.type == "...", then: user_id))` — more accurate than event counts for conversion rates	Critical	Funnel Design
31	Calculate step conversion as `step_N / step_N-1 * 100`	`round(toDouble(step2) / toDouble(step1) * 100, decimals: 1)`	Critical	Metrics

#	Best Practice	Recommended Setting/Value	Priority	Category
32	Calculate overall conversion as `last_step / first_step * 100`	`round(toDouble(step5) / toDouble(step1) * 100, decimals: 2)`	Critical	Metrics
33	Identify checkout abandonment with per-user aggregation	`summarize by:{user_id}` then `filter started_checkout > 0 AND completed_payment == 0`	Recommended	Analysis
34	Measure time between funnel steps	Convert timestamps to millis, subtract, divide by 1000 for seconds: `(unixMillisFromTimestamp(step2_time) - unixMillisFromTimestamp(step1_time)) / 1000.0`	Recommended	Analysis
35	Report `avg`, `median`, and `p95` for inter-step timing	All three are needed to understand the distribution of user behavior	Recommended	Analysis
36	Segment funnels by `event.provider` and `event.category`	Reveals which channels and business lines convert best	Recommended	Segmentation
37	Track daily conversion rate trends over 7+ days	Use `bin(timestamp, 24h)` with `summarize by:{day}` to measure campaign/release impact (`1d` is rewritten to `24h` with a notification; days are UTC-aligned)	Recommended	Trending

5. Revenue and Impact Analysis

#	Best Practice	Recommended Setting/Value	Priority	Category
38	Correlate detected problems with business event volume	Overlay `fetch dt.davis.problems` timelines with `fetch bizevents` volume to visualize impact	Critical	Incident Impact
39	Use `spread:` for concurrent problem timelines	`makeTimeseries count = count(), spread: timeframe(from: event.start, to: coalesce(event.end, now()))`	Recommended	DQL Pattern
40	Compare incident periods against baselines	Compare same hour yesterday or same day last week using `bin(now(), 24h)` offsets	Critical	Incident Impact

#	Best Practice	Recommended Setting/Value	Priority	Category
41	Filter to business hours for impact assessment	Pass your timezone, or 9–5 is UTC: `getDayOfWeek(timestamp, timezone: "<tz>") <= 5 AND getHour(timestamp, timezone: "<tz>") >= 9 AND getHour(timestamp, timezone: "<tz>") < 17` (ISO days: 1 = Monday)	Recommended	Context
42	Convert Dynatrace Intelligence `resolved_problem_duration` from nanoseconds to hours	`resolved_problem_duration / 1h` — the field is in nanoseconds, not milliseconds	Critical	Data Conversion
43	Exclude duplicate problems from impact analysis	`filter dt.davis.is_duplicate == false` . Do not add `dt.davis.is_frequent_event` — frequent issue detection is being phased out and the flag no longer separates anything (ADOPT-03 §5)	Critical	Data Quality
44	Exclude maintenance windows from SLA calculations	`filter maintenance.is_under_maintenance == false`	Recommended	Data Quality
45	Use `sum(toDouble(amount))` for revenue calculations	Always cast `amount` to double before aggregation	Critical	DQL Pattern
46	Track hourly revenue to detect incident-correlated dips	`makeTimeseries hourly_revenue = sum(toDouble(amount)), interval:1h`	Recommended	Monitoring

6. KPIs and Metric Extraction

#	Best Practice	Recommended Setting/Value	Priority	Category
47	Define KPIs that are Specific, Measurable, Actionable, and Time-bound	Every KPI must answer a concrete business question with a formula	Critical	KPI Design
48	Track these five core KPIs at minimum	Transaction Volume (`count()`), AOV (`avg(amount)`), Conversion Rate, Error Rate, Revenue per Hour (`sum(amount)` per hour)	Critical	KPI Design
49	Distinguish business error rates from HTTP error rates	A payment can fail (business error) with HTTP 200 — use `event.type` to identify business failures, not HTTP status	Critical	KPI Design

#	Best Practice	Recommended Setting/Value	Priority	Category
50	Require minimum sample size before calculating error rates	`filter total > 10` before computing error percentages to avoid misleading rates	Recommended	Data Quality
51	Extract critical KPIs as Dynatrace metrics via OpenPipeline	Counter metric / Value metric processors in the OpenPipeline **Metric extraction** stage for order counts and revenue values	Critical	Metric Extraction
52	Set metric dimensions to low-cardinality fields only	Dimensions: `event.provider`, `product_category` — never `order_id` or `user_id`	Critical	Metric Extraction
53	Use extracted metrics for alerting and SLOs	Extracted metrics enable native Dynatrace Intelligence alerting and SLO definitions — DQL on bizevents does not	Critical	Alerting
54	Build a composite business health score	Weight: Transaction Volume 30%, Error Rate 30%, AOV 20%, Conversion Rate 20% — thresholds: Green/Yellow/Red	Recommended	Health Scoring
55	Health score thresholds: Error Rate	Green: < 1%, Yellow: 1–5%, Red: > 5%	Recommended	Health Scoring
56	Health score thresholds: Volume vs Baseline	Green: > 90%, Yellow: 70–90%, Red: < 70%	Recommended	Health Scoring

7. Executive Reporting and Dashboards

#	Best Practice	Recommended Setting/Value	Priority	Category
57	Lead with business outcomes, not infrastructure metrics	Report transactions, revenue, conversion rates — never CPU or memory to executives	Critical	Presentation
58	Always provide baseline context	Compare against yesterday, last week, or last month — never show a number without comparison	Critical	Presentation
59	Quantify incidents in business terms	"Lost 1,200 transactions and $47,000 revenue" not "Service had 500ms latency for 2 hours"	Critical	Presentation
60	One KPI per dashboard tile	Each tile answers exactly one question — no multi-metric tiles	Recommended	Dashboard Design
61	Include a data freshness indicator	`if(latest_event > now() - 5m, then: "CURRENT", else: "STALE")` on every executive dashboard	Recommended	Dashboard Design
62	Executive dashboard must answer three questions	1) Is the business running? 2) Are there problems? 3) What is the trend?	Critical	Dashboard Design

#	Best Practice	Recommended Setting/Value	Priority	Category
63	Automate report delivery via Workflows	Schedule daily reports at 8 AM weekdays with a Time trigger: `0 8 * * 1-5`	Critical	Automation
64	Use `bin(now(), 24h) - 24h` to `bin(now(), 24h)` for "yesterday" queries	Produces clean 24-hour boundaries for scheduled reports	Recommended	DQL Pattern
65	Include a weekly business review query set	Daily volume trend, problems resolved with MTTR, week-over-week volume comparison, busiest hours	Recommended	Reporting
66	Dashboard tile for current transaction rate	`count() / 15.0` over `from:-15m` = transactions per minute	Recommended	Dashboard Design
67	Limit top-N lists to 10 items max in executive views	`limit 10` on all ranked queries for readability	Optional	Dashboard Design

8. DQL Query Patterns

#	Best Practice	Recommended Setting/Value	Priority	Category
68	Always specify a time range on `fetch bizevents`	`fetch bizevents, from:-24h` — never omit the time range	Critical	Performance
69	Filter early, sort late, limit last	Place `filter` immediately after `fetch`, `sort` after `summarize`, `limit` at the end	Critical	Performance
70	Use `==` for exact matches, `~` only for wildcards	`filter event.type == "com.myapp.order.completed"` not `event.type ~ "*order*"`	Critical	Performance
71	Use `in()` with curly-brace arrays for multi-value filters	`filter in(event.type, {"type1", "type2"})` — never SQL-style `IN ('a','b')`	Critical	Syntax
72	Alias all aggregations used in sort	`summarize c = count()` then `sort c desc` — never `sort count() desc`	Critical	Syntax
72b	Wrap multiple `summarize` aggregations in `{}` when using `by:`	With `by:` and 2+ aggregations, wrap them: `summarize { a = count(), b = avg(x) }, by: {field}`. Unwrapped, the tenant emits a `PARAMETERS_SHOULD_BE_GROUPED` INFO (the query still runs). Without `by:`, both forms are canonical and no INFO is emitted.	**Recommended**	Syntax

#	Best Practice	Recommended Setting/Value	Priority	Category
73	Use `round(..., decimals: N)` with named parameter	`round(value, decimals: 2)` — never `round(value, 2)`	Critical	Syntax
74	Use `if(cond, then: ..., else: ...)` with named parameters	Named `then:` and `else:` parameters are mandatory	Critical	Syntax
75	Cast `amount` fields with `toDouble()` before math	`sum(toDouble(amount))` — payload fields may arrive as strings	Recommended	Data Types
76	Use `isNotNull()` before aggregating optional fields	`filter isNotNull(amount)` before `sum(toDouble(amount))`	Critical	Data Quality
77	Use `countDistinctApprox` over `countDistinct` for large datasets	Faster with negligible accuracy loss for >10,000 distinct values	Recommended	Performance
78	Target specific Grail buckets when possible	`fetch bizevents, bucket: {"my_bizevents_*"}` to reduce scan volume	Optional	Performance

9. SLA and SLO Tracking

#	Best Practice	Recommended Setting/Value	Priority	Category
79	Define a Transaction Success Rate SLO	Target: 99.5% — Formula: `successful / total * 100` from payment events	Critical	SLO
80	Monitor business-event ingestion continuity — do not report it as availability	Count empty hours (no business events) in the periods the business transacts; an empty hour means a quiet period or broken capture, not downtime. Availability comes from the success-rate SLO (#79)	Recommended	SLO
81	Define an MTTR SLO	Target: < 2 hours — Formula: `avg(resolved_problem_duration)` in hours, excluding duplicates and maintenance-window problems (#44)	Recommended	SLO
82	Define a Revenue Impact per Incident SLO	Target: < $5,000 — Formula: baseline revenue minus incident-period revenue	Recommended	SLO

#	Best Practice	Recommended Setting/Value	Priority	Category
83	Do not derive SLA availability from summed problem durations	Problems overlap, so summed durations are entity-hours of impact, not downtime — they can exceed the period many times over (BIZEV-04 §6). Define availability as an SLO on good ÷ total events (SLO-02/03)	Critical	SLA
84	Include SLO status indicator in every report	`if(success_rate >= 99.5, then: "MET", else: "MISSED")`	Critical	Reporting
85	Track MTTR with `avg`, `median`, and `p95`	Report all three — avg shows the mean, median shows the typical case, p95 shows worst cases	Recommended	Metrics

10. Gen2 vs Gen3 Adoption

Practices from BIZEV-07 for environments that have not (fully) moved to the Gen3 experience.

#	Best Practice	Recommended Setting/Value	Priority	Category
86	Confirm the platform generation before designing business analytics	Dynatrace Managed = no business events in any form (classic toolchain only); SaaS with Grail = adopt business events, regardless of how classic the day-to-day operation is	Critical	Adoption
87	On Grail-enabled SaaS tenants, build new business analytics on business events, not classic mechanisms	Calculated service metrics are closed to new customers and superseded by OpenPipeline metric extraction — new classic investment is migration debt	Critical	Adoption
88	Use the minimal Gen3 footprint for hybrid adoption	Capture via Settings (OneAgent rules — Full-Stack mode required), report in Notebooks + the new Dashboards app, grant `storage:bizevents:read`; add OpenPipeline only if you need custom buckets or new `bizevents.*` metrics (the classic pipeline is not available to accounts created from September 2026) — nothing else has to move	Recommended	Adoption
89	Bridge business KPIs to classic dashboards with metric extraction	`bizevents.`-prefixed extracted metrics are readable in Data Explorer, classic dashboards, and classic metric-event alerting (classic pipeline: not available to accounts created from September 2026 and to be deactivated in a future release — build new extraction in OpenPipeline)	Recommended	Adoption

#	Best Practice	Recommended Setting/Value	Priority	Category
90	Create metric-extraction rules at capture time, not reporting time	Extraction is forward-only — no backfill; with classic-pipeline extraction, late-timestamped events land in a `.failed`-suffixed key	Critical	Adoption
91	Capture must-not-miss events server-side, not (only) via RUM	`sendBizEvent` is dropped for unmonitored sessions (cost and traffic controls) — orders and payments belong on OneAgent capture rules or the ingest API	Critical	Adoption
92	On Managed, apply reverse-domain naming to classic artifacts	Name request attributes, session properties, and log metrics in `com.<company>.<domain>.<action>` style so the eventual migration is a mapping exercise	Recommended	Adoption
93	Size business events against your actual license model before rollout	Classic DDU: 100 DDUs/GB ingest, 0.30 DDUs/GB-day retain, 1.70 DDUs/GB query, 200,000 DDUs/year free tier; DPS: Events powered by Grail capability	Recommended	Adoption
94	Scope business-analyst access with a bucket-limited IAM policy	`ALLOW storage:bizevents:read WHERE storage:bucket-name = "default_bizevents";` plus the matching `storage:buckets:read` statement — grants Gen3-app access to business events only, nothing else in Grail; route sensitive streams to custom buckets for per-team policies	Recommended	Adoption

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